

Presence Sensing Lights Controller

The 'Presence Sensing Lights Controller' construction project, published in Nov. 2012 issue, was a good project. I am interested in this project but I have some questions and doubts. How to write the program in Keil μ Vision4 compiler? Where can I get a suitable programmer and its tutorials? Please help me.

Ramakanta Mohanta
Keonjhar, Odisha

The author Sanjib Bhuiya replies:

First of all, I thank you for your interest in this circuit. Answers to your queries can be found on the website links provided below:

<http://www.circuitstoday.com/getting-started-with-keil-uvision>

<http://www.slideshare.net/anish-goel/keil-tutorial>

Cellphone-based Remote Controller for Water Pump

I have some doubts in 'Cellphone-based Remote Controller for Water Pump' circuit published in January issue. Can you please provide the detailed working of DTMF IC? How can I activate cellphone in 'auto answering' mode?

Neha Jaiswal

Shri Guru Gobind Singhji Institute of Engg & Tech, Nanded (Maharashtra)

The author Abdul Nasir replies:

For detailed working of MT8870 DTMF IC, refer to the application note of MT8870 at '<http://www.zarlink.com/zarlink/msan108-appnote.pdf>'

If you want to know more about DTMF decoding, please refer to '<http://www.dattalo.com/technical/theory/dtmf.html>'

The 'auto answering' facility can be activated using headset settings. In case of a Nokia mobile phone, this can be done as follows: Menu→Settings→Enhancements→Headset→Automatic answer→On

GSM-based Borewell Water-Level Monitor

Regarding 'GSM-based Borewell Water-Level Monitor' construction

project published in May 2012 issue, I have following queries:

1. What is the mobile number included in the code?

2. The 'AT+CMGS' command is not used anywhere in the code. Please check whether the code is error-free.

3. What should be the output of transistor T1 in volts when A and B sensors are not in contact (out of water) and when both of these sensors are immersed in water?

4. Is it necessary to complete the entire circuit for the microcontroller's output pin 3 (P1.2) status to change?

5. In the initial part of our testing, the T1 collector output was 2.68V when A and B were open (out of water), and 5V when A and B were closed (immersed in water).

However, when we had connected the collector of T1 to pin 12 of the microcontroller, and A and B were open, the output suddenly became 3.5V. When A and B were closed, the output became 5V. What could be the reason?

When A and B are open, T1 collector voltage should be less than 0.8V. The TTL logic levels correspond to 0-0.8V=logic 0 and 2.5V-5V=logic 1. Hence for our circuit to work properly, the output of T1 should be less than 0.8V (not 2.8V) when A and B are open.

6. Another problem is that the microcontroller output (P1.2) remained the same when A and B were open or close.

Jaideep Mehendale

Through e-mail

The author Gurunath Reddy replies:

1. The mobile number included in the code is the SIM number used by the farmer. If water level dips below the threshold level in the borewell, he will get a ring in his cell phone. Motor pump can be switched off from any other mobile phone by sending 'motor11 off' message to the SIM in the modem.

2. The program code is correct. AT+CMGS is not used in the code because the GSM modem is not sending any message. GSM modem gives only a ring to the number used in the code.

'Spot An Error' Award Winners

On 'Contents' page of March issue, 'Soldering Iron Tip Preserver' is mistakenly printed under 'Circuit Ideas' list. This article was published in February issue.

Adil Khan

□ The polarity of capacitor C6 in 'SMS-based Automation System' and C1 in '31-channel RF Remote Control' is reverse printed.

Sankara Narayanan

□ In the article 'Digital Storage Oscilloscopes,' harmonic waveforms on page 58 are wrongly mentioned as 2nd and 3rd harmonics in place of 3rd and 5th, respectively.

Vasudevan Nambudiri

3. In the circuit, the output of transistor is 'low' when sensors (A and B) are not in contact with water and it goes 'high' when sensors are in contact with water. Since I have used edge-triggered interrupt facility available in 8051, it doesn't matter whether the output of the transistor (T1) is high or low.

We only need level change—either from '0' to '1', or from '1' to '0'. Make sure that when water comes in contact with sensors, there is corresponding change in the voltage at the output of the transistor. Remember only edge-triggering at the input pin of 8051 can run the call interrupt routine. Also make sure that the output of transistor is in TTL logic. You can also use a NOT gate instead of transistor T1 to get perfect logic transitions.

4. Logic at pin P1.2 changes only when a proper SMS is received through the mobile phone. The entire circuit need not be completed. If the microcontroller is executing interrupt routine, it is not possible to change the motor status because the microcontroller gives first priority to the interrupt rather than sequential execution.

5. Try using a higher value of resistor near the transistor until you get the desired TTL logic. The TTL logic levels correspond to 0-0.8V=logic 0 and 2.5V-5V=logic 1.

6. Microcontroller output (P1.2), which is connected to the motor,